

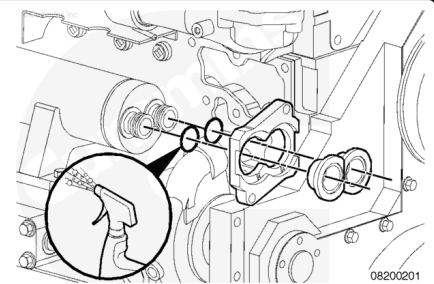
Trainings ▾

Install

With EGR

⚠ CAUTION ⚠

Do not use petroleum products on the coolant plumbing o-rings. The o-rings will swell and cause the connection to leak.



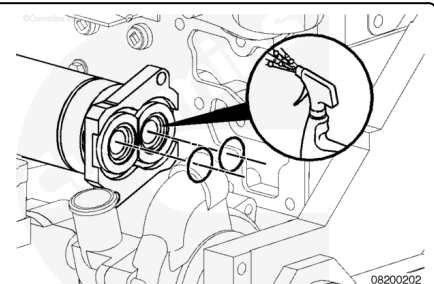
To avoid pinching or rolling the o-ring during assembly, do **not** lubricate the o-ring. Lubricate the o-ring **connection mating surfaces** with clean engine coolant, soapy water, or vegetable oil.

Install new o-rings on the lubricating oil cooler water connection tubes.

While holding the connection flange in place, install the water connections into the lubricating oil cooler. The straight edges of the connection **must** face each other.

⚠ CAUTION ⚠

Do not use petroleum products on the coolant plumbing o-rings. The o-rings will swell and cause the connection to leak.



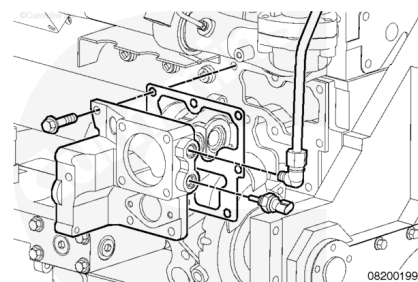
To avoid pinching or rolling the o-ring during assembly, do **not** lubricate the o-ring. Lubricate the o-ring **connection mating surfaces** with clean engine coolant, soapy water, or vegetable oil.

Install two new o-rings on the end of the lubricating oil cooler water tube connections.

Use a new gasket, and install the thermostat housing support and capscrews.

Torque Value: 54 n•m [40 ft-lb]

Install the coolant temperature sensor. Refer to Procedure 019-019 in Section 19 (</qs3/pubsys2/xml/en/procedures/88/88-019-019.html>) in the Troubleshooting and Repair Manual, CM870 Electronic Control System, ISM Engines, Bulletin 4021381.

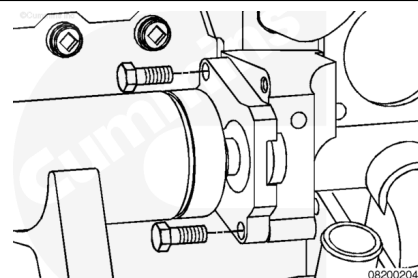


Install the air compressor coolant outline line. Refer to Procedure 012-109 in Section 12. (</qs3/pubsys2/xml/en/procedures/35/35-012-109-tr.html>)

Install the two oil cooler water connection retainer mounting flange capscrews.

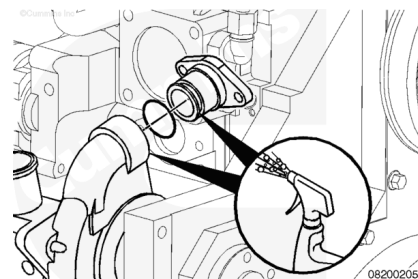
Torque Value: 47 n•m [35 ft-lb]

Make sure the o-rings on the end of the connection tubes are located correctly.



⚠ CAUTION ⚠

Do not use petroleum products on the coolant plumbing o-rings. The o-rings will swell and cause the connection to leak.

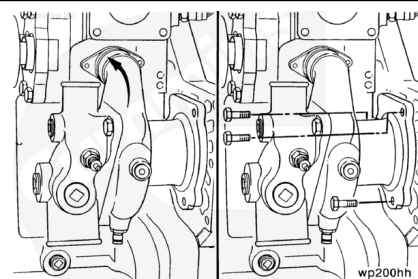


To avoid pinching or rolling the o-ring during assembly, do **not** lubricate the o-ring. Lubricate the o-ring **connection mating surfaces** with clean engine coolant, soapy water, or vegetable oil.

Install a new o-ring on the water pump water transfer tube.

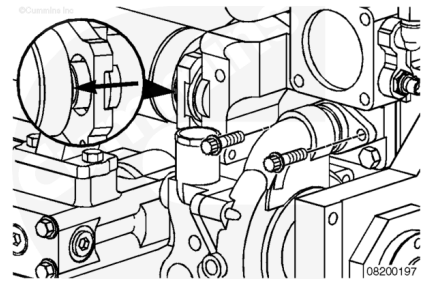
Install the water transfer tube and o-ring assembly in the water pump body.

Install the water pump. Refer to Procedure 008-062 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-062-tr.html>)



Use a new gasket, and install the water pump water transfer tube mounting capscrews.

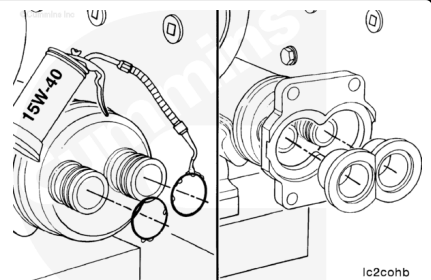
Torque Value: 24 n•m [212 in-lb]



Without EGR

⚠ CAUTION ⚠

Do not use petroleum products on the coolant plumbing o-rings. The o-rings will swell and cause the connection to leak.



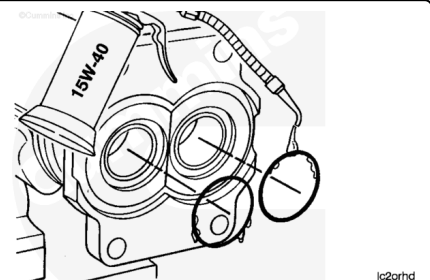
To avoid pinching or rolling the o-ring during assembly, do **not** lubricate the o-ring. Lubricate the o-ring **connection mating surfaces** with clean engine coolant, soapy water, or vegetable oil.

Install new o-rings on the lubricating oil cooler water connection tubes.

While holding the connection flange in place, install the water connections into the lubricating oil cooler. The straight edges of the connection **must** face each other.

⚠ CAUTION ⚠

Do not use petroleum products on the coolant plumbing o-rings. The o-rings will swell and cause the connection to leak.



To avoid pinching or rolling the o-ring during assembly, do **not** lubricate the o-ring. Lubricate the o-ring **connection mating surfaces** with clean engine coolant, soapy water, or vegetable oil.

Install two new o-rings on the end of the lubricating oil cooler water tube connections.

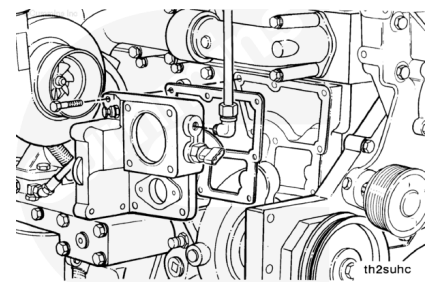
Use a new gasket, and install the thermostat housing support and capscrews.

Tighten the capscrews.

Torque Value: 54 n•m [40 ft-lb]

Install the coolant temperature sensor.

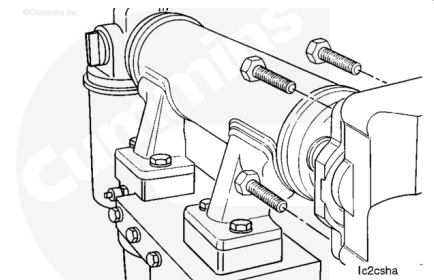
Install the air compressor coolant supply line.



Install the four mounting flange capscrews. Alternately and evenly, tighten the four capscrews.

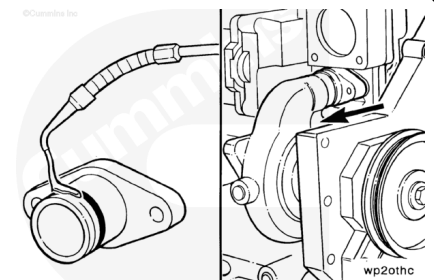
Torque Value: 47 n•m [35 ft-lb]

Be sure the o-rings on the end of the connection tubes are located correctly.



⚠ CAUTION ⚠

Do not use petroleum products on the coolant plumbing o-rings. The o-rings will swell and cause the connection to leak.



To avoid pinching or rolling the o-ring during assembly, do **not** lubricate the o-ring. Lubricate the o-ring **connection mating surfaces** with clean engine coolant, soapy water, or vegetable oil.

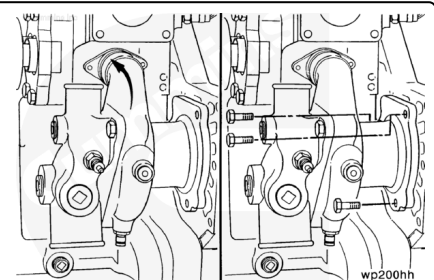
Install a new o-ring on the water pump water transfer tube.

Install the water transfer tube and o-ring assembly in the water pump body.

Rotate the water pump inward and install the three water pump mounting capscrews.

Tighten the capscrews.

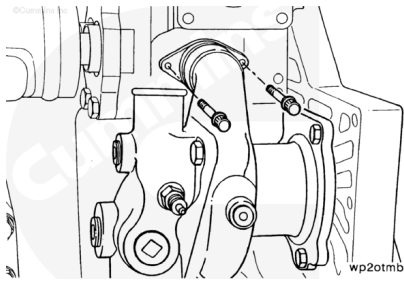
Torque Value: 47 n•m [35 ft-lb]



Use a new gasket and install the water pump water transfer tube mounting capscrews.

Tighten the capscrews.

Torque Value: 24 n•m [212 in-lb]



Last Modified: 01-Apr-2015
